

Greedy Notes

LeetCode 561: Array Partition

Pattern: Greedy + Sorting

 Bloomberg

 Amazon

★ Key Observation

The problem asks us to pair up array elements such that the sum of the minimum of each pair is maximized.

We want to maximize:

$$\min(\text{pair1}) + \min(\text{pair2}) + \dots + \min(\text{pairN})$$

If we sort the array, the elements satisfy $a \leq b \leq c \leq d \leq \dots$

To prevent larger numbers from being "wasted" by being paired with much smaller numbers, we should **always pair adjacent elements** together.

Sorted Array:

a	b	c	d
Pair 1		Pair 2	

Answer: The sum will be $a + c + e + \dots$

i.e., every alternate element starting from index 0.

C++ Implementation

```
class Solution {  
public:  
    int arrayPairSum(vector<int>& nums) {  
        sort(nums.begin(), nums.end());  
        int n = nums.size();  
        int ans = 0;  
  
        for(int i = 0; i < n; i += 2) {  
            ans += nums[i];  
        }  
  
        return ans;  
    }  
};
```

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